Statistics

The big idea

Statistics is about collecting, organising and **presenting** data so that patterns are easy to see. This chapter focuses on graphs: **bar graphs**, **histograms** and **frequency polygons**.

Organising data

Raw data is the unorganised list of observations. A **frequency distribution table** groups it: each value or **class interval** with its **frequency** (how many times it occurs). In 10–20, 10 is the lower class limit and 20 the upper; the **class size** is 10 and the **class mark** is 15. By convention, a value of 20 goes into 20–30, not 10–20.

Bar graphs

Rectangular bars of **equal width** with **equal gaps**; the height shows the value. Used for discrete categories (e.g. favourite sports, monthly expenditure).

Histograms

A histogram represents **continuous** grouped data. The bars are drawn on the class intervals **with no gaps**.

UNEQUAL CLASS WIDTHS

If class widths differ, the bar heights must be adjusted so that **area** represents frequency:
adjusted height = (minimum class size ÷ class size) × frequency.

KINK ON THE AXIS

If the first class starts far from 0 (e.g. 150–155 cm), draw a small zig-zag (kink) near the origin to show a break in the scale.

Frequency polygons

Join the **midpoints of the tops** of histogram bars with straight lines. To close the shape, add a class with **zero frequency** before the first class and after the last, and join to their midpoints. You can also draw a frequency polygon directly by plotting (class mark, frequency).

READING A FREQUENCY TABLE

Marks: 0–10: 5, 10–20: 10, 20–30: 4, 30–40: 6, 40–50: 7.
Class marks: 5, 15, 25, 35, 45. The polygon joins (5, 5), (15, 10), (25, 4), (35, 6), (45, 7), with ends at (–5, 0) and (55, 0).

Quick recap

- Frequency tables organise raw data.
- Bar graph: separated bars. Histogram: touching bars, area ∝ frequency.
- Frequency polygon: join midpoints of bar tops (or plot class marks).

Practice questions

1. The class mark of 30–40 is
   (a) 30 (b) 35 (c) 40 (d) 70

2. In a histogram, the bars
   (a) have gaps between them (b) touch each other (c) are circles (d) are drawn horizontally only

3. The class size of the interval 150–155 is
   (a) 150 (b) 5 (c) 155 (d) 152.5

4. A frequency polygon is obtained by joining the
   (a) midpoints of the tops of the bars in a histogram (b) corners of the bars (c) lower limits of classes (d) cumulative frequencies

5. A value of 20 in classes 10–20 and 20–30 is placed in
   (a) 10–20 (b) 20–30 (c) both (d) neither

Answers: 1. b 2. b 3. b 4. a 5. b